Bikeshare Rebalancing Problem: Final MILP Model Proposal

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1. Introduction

This document presents the revised Mixed-Integer Linear Programming (MILP) model for the bike-sharing rebalancing problem, incorporating the feedback provided by Professor Lenz. The model captures the redistribution of bikes among stations over discrete time steps, minimizing total transportation, holding, and unmet demand costs.

2. Sets and Indices

- S: set of stations, indexed by i, j
- T: set of time periods $t = 1, \ldots, T$
- K: (optional) set of available trucks $k = 1, \ldots, K$

3. Parameters

- $D_{i,t}$: demand at station i during period t
- $I_{i,0}$: initial number of bikes at station i
- C_i: capacity of station i
- $c_{i,j}$: transportation cost (e.g., distance or time) between stations i and j
- h: holding cost per bike remaining at a station
- p: penalty cost for unmet demand
- F: maximum number of trucks available (if capacity constraint added)

4. Decision Variables

- $f_{i,j,t} \geq 0$: number of bikes transported from station i to station j at time t
- $I_{i,t} \geq 0$: number of bikes available at station i at the end of period t
- $B_{i,t} \geq 0$: unmet demand (slack) at station i during period t
- (Optional) $x_{i,j,t} \in \{0, 1\}$: binary variable indicating whether a truck travels from i to j at time t

5. Objective Function

$$\min \; Z = \sum_{t \in T} \sum_{\substack{i,j \in S \\ i \neq j}} c_{i,j} f_{i,j,t} + h \sum_{t \in T} \sum_{i \in S} I_{i,t} + p \sum_{t \in T} \sum_{i \in S} B_{i,t}$$

Minimize the total transportation, holding, and penalty costs.

6. Constraints

6.1 Bike Balance at Each Station and Time Step

$$I_{i,t} = I_{i,t-1} + \sum_{\substack{j \in S \\ j \neq i}} f_{j,i,t} - \sum_{\substack{j \in S \\ j \neq i}} f_{i,j,t} - D_{i,t} + B_{i,t} \quad \forall i \in S, \forall t \in T$$

6.2 Station Capacity

$$0 \leq I_{i,t} \leq C_i \quad \forall i \in S, \forall t \in T$$

6.3 Non-Negativity

$$f_{i,j,t}, I_{i,t}, B_{i,t} \geq 0$$

6.4 (Optional) Fleet-Size Constraint

$$\sum_{\substack{i,j \in S \\ i \neq j}} x_{i,j,t} \leq F \quad \forall t \in T$$

and

$$f_{i,j,t} \leq M \cdot x_{i,j,t}$$

where M is a large constant linking flow and binary decisions.

7. Model Discussion

The variable $B_{i,t}$ acts as a slack variable representing unmet demand, penalized in the objective function. The term $h \cdot I_{i,t}$ represents a small holding cost to prevent excessive bike accumulation.

The current version assumes unlimited truck availability, but this can be restricted via the optional fleet-size constraint or analyzed in postprocessing.

8. Possible Model Extensions

- **Multiple trucks and routing:** Extend the model with an additional truck index k and routing constraints.

- **Travel times and lead times:** Incorporate time delays between stations based on distances.
- **Service-level constraints:** Ensure that each station satisfies a minimum demand percentage (e.g., 90%).
- **Demand uncertainty:** Represent $D_{i,t}$ as a random variable or use scenario-based optimization.
- **Sustainability aspects:** Include battery-electric truck limitations or emission constraints.

9. Next Steps

The next version will incorporate the above adjustments and be tested on the real Capital Bikeshare dataset to evaluate the performance and interpret operational insights.

Best regards,
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